# Arman Acharya

B.Tech CSE (Data Science) · MLR Institute of Technology · Hyderabad, India

25r21a6704@mlrit.ac.in · GitHub: github.com/ACHARYA-SAMA · Portfolio: portfolio-amber-three-dt9gbfu54u.vercel.app

## PROFILE

Computer Science (Data Science) student working where code meets the physical world: cameras that track, circuits that react, interfaces that feel alive. Frontend developer and designer with hands-on computer vision, IoT and hackathon experience. Black belt and state-level karate champion. Open to projects, internships and hackathon teams.

## ACHIEVEMENTS

- **Hackathons:** entered 5+; finished **Top 5 in every one**, with **one first place**.
- Top 5: **Yantra Yugam**, **GDGoC × MREM TechSprint**, **Bower AI**, **MSME Idea Hackathon 6.0**. Finalist: **ZIGNASA 2K25**.
- **Smart India Hackathon (ISRO problem statement):** cleared both internal and external campus evaluation rounds; built an AI-based virtual camera tracking system using an Extended Kalman Filter.
- **Black Belt & State-Level Karate Champion.**

## EXPERIENCE & LEADERSHIP

**Frontend Web Developer** — SQUAD, CSE Data Science club, MLRIT

- Built the club's frontend repository and maintain it.
- Live: squad-mlrit.vercel.app · squad-web-rouge.vercel.app · squad-club-web.vercel.app

**Lead Designer** — Club Literati, MLRIT

- Vector layouts and promotional flyers for events including Interaction Day and Murder Mystery.

## PROJECTS

**Micro-Sort** · OpenCV, YOLOv8-nano, Raspberry Pi 4

AI-driven 4-axis SCARA sorting robot for MSMEs; fully offline computer vision on a Raspberry Pi 4.

**IoT Seatbelt Safety System** · Arduino, IoT, Embedded

Arduino-based automotive safety hardware tying seatbelt status to vehicle behaviour.

**Advanced Recommendation Systems Pipeline** · Python, NumPy

Candidate retrieval plus ranking using cosine similarity, matrix factorisation and neural collaborative filtering.

**UrbanFlow Smart Traffic** · React, Vite

Intelligent traffic management with real-time simulation and an interactive dashboard showing AI-driven signal decisions.
GitHub · Live (Vercel)

**Civil Aviation India** · Next.js, three.js, React, Tailwind

Website for the Aero Space Society of India and the Indian Institute of Aero Space Technology and Management.
GitHub · Live (Vercel)

**Scyte** · Next.js, TypeScript

Studio website and product platform (web, mobile, UI/UX, AI-powered solutions).
GitHub · Live (Vercel)

**HAY** · Next.js, Tailwind CSS

Ticket booking app.
GitHub · Live (Vercel)

**PricePilot** · Python, FastAPI, NumPy, Pandas, React

Local event revenue optimizer with least-squares demand modeling, Newton-Raphson optimization and diagnostics.
GitHub

**BITL** · Python, MediaPipe, OpenCV

Gesture-controlled AR playground for the webcam: neon air drawing, blocks and Jujutsu domains.
GitHub

**RADIX LedgerLens** · React, Vite, Express
Financial analytics and voice assistant.
GitHub

**RADIX FSOC Coarse-Alignment Simulator** · Python, PyQt6, NumPy, OpenCV
Desktop app for AI-assisted coarse alignment of mobile free-space optical communication terminals.
GitHub

**Portfolio** · Next.js, Three.js
Interactive racing-themed personal portfolio.
GitHub · Live (Vercel)

More repositories: scyte1, scyte, Squad-web, squad-mlrit-web, squad-mlritMK1, civiplaviationindiasatyanaryna.

## SKILLS

**Web:** Next.js, React, Tailwind CSS, Three.js · **Languages:** Python, Java, C · **AI & Data:** Computer Vision (OpenCV), NumPy, Recommendation Systems, Collaborative Filtering, Neural Networks · **Hardware & IoT:** Arduino, Raspberry Pi · **Design:** Figma, Canva

## EDUCATION

**B.Tech, Computer Science & Engineering (Data Science)** — MLR Institute of Technology, Hyderabad